FermatQuinticJacobian: Lean 4 Certificate for the Jacobian Smoothness of the Fermat Quintic Fourfold

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/FermatQuinticJacobian
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23652994

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Abstract

Standalone Lean 4 formalization (‘FermatQuinticJacobian.lean‘) certifying the non-degeneracy of the partial derivatives and Jacobian smoothness criterion for the Fermat quintic fourfold V(x_0^5 + ... + x_5^5) subset P^5.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Certifies Jacobian smoothness of the Fermat quintic host variety in Lean 4. Explicitly states that Jacobian smoothness is a prerequisite geometric property and not a resolution of the Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/FermatQuinticJacobian`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652994`
- **Earliest Git Commit SHA:** `777a2e97451ec4a366c03bbacb4333c001bb833d` (2026-10-03 12:49:57 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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A Lean 4 certificate, checked against Mathlib ‘v4.22.0‘, that the Fermat quintic
“‘text
G = X0^5 + X1^5 + X2^5 + X3^5 + X4^5 + X5^5
“‘
has partial derivatives
“‘text
pderiv i G = 5 * X i ^ 4
“‘
Those forms vanish together only at the origin of affine 6-space. By the Jacobian criterion, ‘V(G)‘ in ‘P^5‘ is smooth.
The theorem in the repository is ‘Hodge.FermatQuintic.partial_eq‘.
This is not a proof of the Hodge conjecture. It does not construct a cycle from a

Hodge class. It does not show that ‘H^{3,1}‘ vanishes: on this fourfold that primitive piece is the degree-4 part of ‘C[X_i]/(X_i^4)‘, of dimension 120. The non-homogeneous expression ‘sum x_i^5 - 5 x0 x1 x2 x3 x4 x5‘ is not used. The product of six variables has

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/FermatQuinticJacobian` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`FermatQuinticJacobian.lean`	1871 B	`d94c40a67456932d...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	638 B	`1d6c88adb668f721...`
`README.md`	1106 B	`775cf9e9962b7aa2...`
`lakefile.toml`	288 B	`3f8f4cf04ccf5e35...`
`lean-toolchain`	25 B	`8f6680f8389e8ade...`

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A Verification certificate

The formal development was machine-checked as follows.

- Repository: `https://github.com/BenFrohman/FermatQuinticJacobian`
- Verified commit: `8ca201272348ac9047601ea60e4567d3e2c984a9`
- Lean toolchain: `leanprover/lean4:v4.22.0`
- Mathlib: tag `v4.22.0`, commit (from `lake-manifest.json`) `79e94a093aff4a60fb1b1f92d9681e407124c2ca`
- Continuous integration run (build, axiom audit, `sorry`/`axiom` scan; passed): `https://github.com/BenFrohman/FermatQuinticJacobian/actions/runs/41568055306`
- SHA-256 of the GitHub source tarball of the verified commit, downloaded twice with identical result: `https://github.com/BenFrohman/FermatQuinticJacobian/archive/8ca201272348ac9047601ea60e4567d3e2c984a9.tar.gz` `c03d038daf33886f9f0e87cd8d9e2a49e937ff88ba91dc175be345e6939c0aa6`

The tarball hash refers to the verified commit rather than to a release tag, since a document cannot contain the hash of an archive that includes itself. The Lean sources are unchanged between that commit and the release tag.

The development contains no `sorry` and no user-declared axioms. Every one of the 3 theorems and lemmas listed below depends only on Lean’s standard axioms `propext`, `Classical.choice` and `Quot.sound` (or on none). The table is the verbatim output of `lake env lean AxiomAudit.lean` (`#print axioms`) on the verified commit.

Lean declaration	`#print axioms`
`Hodge.FermatQuintic.partial_eq`	`[propext, Classical.choice, Quot.sound]`
`Hodge.FermatQuintic.partial_off`	`[propext, Classical.choice, Quot.sound]`
`Hodge.FermatQuintic.partial_on`	`[propext, Classical.choice, Quot.sound]`